

PROBLEM 1

$$y(t) = x^3(t)$$

$$x(t) \rightarrow \boxed{A} \rightarrow y(t) = x^3(t)$$

DEFINE:

$$x_1(t) \rightarrow \boxed{A} \rightarrow y_1(t) = x_1^3(t) = (x_1(t))^3$$

$$x_2(t) \rightarrow \boxed{A} \rightarrow y_2(t) = x_2^3(t)$$

$$x_3(t) = x_1(t) + x_2(t) \rightarrow y_3(t) = (x_1(t) + x_2(t))^3$$

LET'S MAKE THIS MORE COMPLICATED

$$x_3(t) = a x_1(t) + b x_2(t) \rightarrow y_3(t) = (a x_1(t) + b x_2(t))^3$$

$$y_3(t) = a^3 x_1^3(t) + 2a^2 x_1^2(t) b x_2(t) + \dots$$

$$\text{Does } y_3(t) = a y_1(t) + b y_2(t)?$$

$$a y_1(t) + b y_2(t) = a x_1^3(t) + b x_2^3(t)$$

$$y_3(t) \neq a y_1(t) + b y_2(t)$$

$\therefore$ NOT LINEAR.

Problem 2

$$y[n] = 3x[n] - 2$$

$$x[n] \rightarrow \boxed{4} \rightarrow y[n] = 3x[n] - 2$$

Define:

$$x_1[n] \rightarrow y_1[n] = 3x_1[n] - 2$$

$$x_2[n] \rightarrow y_2[n] = 3x_2[n] - 2$$

$$x_3[n] = ax_1[n] + bx_2[n] \rightarrow y_3[n] = 3x_3[n] - 2$$

$$y_3[n] = 3(ax_1[n] + bx_2[n]) - 2$$

$$y_3[n] = 3ax_1[n] + 3bx_2[n] - 2$$

Does ~~$ay_1[n] + by_2[n] = y_3[n]$~~ $ay_1[n] + by_2[n] = y_3[n]$?

$$\begin{aligned} ay_1[n] + by_2[n] &= a(3x_1[n] - 2) + b(3x_2[n] - 2) \\ &= 3ax_1[n] - 2a + 3bx_2[n] - 2b \end{aligned}$$

$$\therefore y_3[n] \neq ay_1[n] + by_2[n]$$

THE SYSTEM IS NON-LINEAR

Problem 3

$$x[n] \rightarrow \boxed{A} \rightarrow y[n] = \operatorname{Re} \{ x[n] \}$$

first LETS CONSIDER

$$x[n] \rightarrow y[n] = x[n]$$

$$x_1[n] \rightarrow y_1[n] = x_1[n]$$

$$x_2[n] \rightarrow y_2[n] = x_2[n]$$

$$x_3[n] = ax_1[n] + bx_2[n] \Rightarrow y_3[n] = ay_1[n] + by_2[n]$$

$$ay_1[n] + by_2[n] = ay_1[n] + by_2[n]$$

$$\therefore y_3[n] = ay_1[n] + by_2[n]$$

$\therefore$ IT IS LINEAR.

... BUT WE NOW HAVE TO CONSIDER THE $\operatorname{Re} \{ x[n] \}$

$$x[n] \rightarrow y[n] = \operatorname{Re} \{ x[n] \}$$

$$x[n] = \underset{\substack{\uparrow \\ \text{REAL}}}{r[n]} + j \underset{\substack{\uparrow \\ \text{IMAGINARY}}}{s[n]} \quad \rightsquigarrow \quad x_1[n] \rightarrow y_1[n] = r_1[n]$$

CHECK SCALING FIRST...

$$x_2[n] = ax_1[n] \quad \text{WHERE } a = j$$

$$\begin{aligned} x_2[n] &= j(r_1[n] + js_1[n]) \\ &= -s_1[n] + jr_1[n] \end{aligned}$$

$$x_2[n] \rightarrow y_2[n] = -s_1[n]$$

$$ay_1[n] = j r_1[n]$$

$$y_2[n] \neq ay_1[n]$$

SCALING IS NOT SATISFIED

$\therefore$ NOT LINEAR.

Problem 4

$$x[n] \rightarrow \boxed{A} \rightarrow y[n] = 3x[n] - 2$$

$$\Rightarrow x_1[n] \rightarrow y_1[n] = 3x_1[n] - 2$$

$$x_2[n] = x_1[n + n_0] \Rightarrow y_2[n] = 3x_1[n + n_0] - 2$$

SHIFT
SO IF A ~~THE~~ OUTPUT BY n_0

$$y_1[n + n_0] = 3x_1[n + n_0] - 2$$

A SHIFT IN THE INPUT EQUALS
THE SAME SHIFT IN THE OUTPUT

∴ TIME INVARIANT

PROBLEM 5

$$X(t) \longrightarrow \boxed{A} \longrightarrow Y(t) = X^3(t)$$

DEFINE:

$$X_1(t) \rightarrow Y_1(t) = X_1^3(t)$$

$$X_2(t) = X_1(t+t_0) \rightarrow Y_2(t) = X_1^3((t+t_0)^2)$$

↑
SOME ARBITRARY
TIME SHIFT

$$Y_1(t) = X_1^3(t^2 + 2tt_0 + t_0^2)$$

IF WE SHIFT $Y_1(t)$ BY THE SAME t_0

$$Y_1(t+t_0) = X_1^3(t^2+t_0)$$

$$Y_2(t) \neq Y_1(t+t_0)$$

A SHIFT APPLIED TO THE INPUT DOESN'T RESULT
IN THE SAME SHIFT TO THE OUTPUT.

∴ TIME VARIANT.

Problem 6

$$y[n] = x[n] - x[n+1]$$

$$x[n] \rightarrow \boxed{A} \rightarrow y[n] = x[n] - x[n+1]$$

DEFINITION:

$$X_1[n] \rightarrow Y_1[n] = X_1[n] - X_1[n+1]$$

$$X_2[n] = X_1[n+n_0] \rightarrow Y_2[n] = X_1[n+n_0] - X_1[n+n_0+1]$$

↑
SOME ARBITRARY
SHIFT IN n

IF WE SHIFT $Y_1[n]$ BY THE SAME n_0

$$Y_1[n+n_0] = X_1[n+n_0] - X_1[n+n_0+1]$$

$$Y_2[n] = Y_1[n+n_0]$$

A SHIFT APPLIED TO THE INPUT RESULTS IN THE SAME THING AS A TIME SHIFT TO THE OUTPUT.

∴ Time INVARIANT